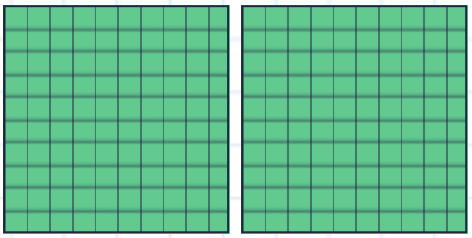
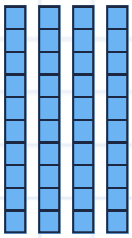
START HERE

Place Value Face Value

A step-by-step guide. Read one page at a time. Look at the blocks. Then try the **Your turn** boxes.



2 hundreds



4 tens



3 ones

Hundreds	Tens	Ones
2	4	3

IN THIS BOOK YOU WILL LEARN

1. Digits 0 to 9
2. Ones
3. 10 ones make a ten
4. 10 tens make a hundred
5. The number house
6. Why the place matters
7. Face value
8. Place value
9. Face value vs place value
10. The Zero Hero
11. Expanded form
12. Practice + answers

My name: _____ **Class:** _____

Digits: our 10 number friends

In maths we have only **10 digits**:

0 **1** **2** **3** **4** **5** **6** **7** **8** **9**

With just these 10 digits we can write **every** number: 7, 45, 382, 999 ...

THINK ABOUT IT

Look at **25** and **52**. They use the **same digits**, 2 and 5.

Are they the same number? **No!** 52 is much bigger.

The **place** where a digit sits changes how much it is worth. This book will show you why.

YOUR TURN

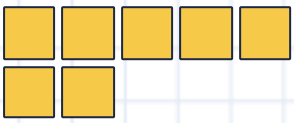
1. How many digits are there? _____
2. What is the biggest digit? _____ The smallest? _____
3. Make two different numbers using the digits 3 and 8: _____ and _____

Ones

1 **This little cube is ONE.**



2 **Count the cubes: 7 cubes = 7 ones.**



7 ones

3 **Keep counting: 10 cubes = 10 ones.**



10 ones

Counting lots of loose cubes is slow. So when we get 10 ones, we do something clever... (turn the page!)

YOUR TURN

How many ones?



_____ ones



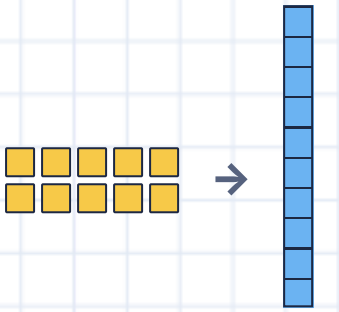
_____ ones

10 ones make 1 ten

1. **Take 10 ones.**



2. **Stick them together into one long rod. This rod is called a TEN.**

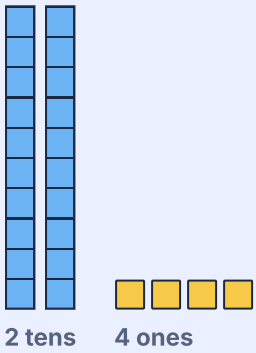


REMEMBER

10 ones = 1 ten

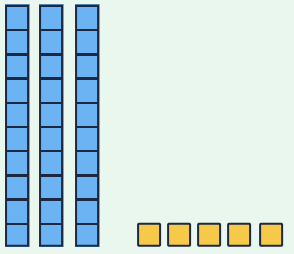
Groups of ten are everywhere: 10 fingers · a packet of 10 biscuits · a ₹10 note · 10 toes.

TENS AND ONES TOGETHER

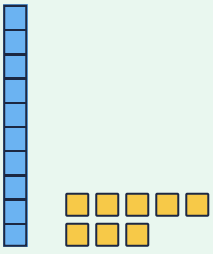


= **2 tens and 4 ones = 24**

YOUR TURN



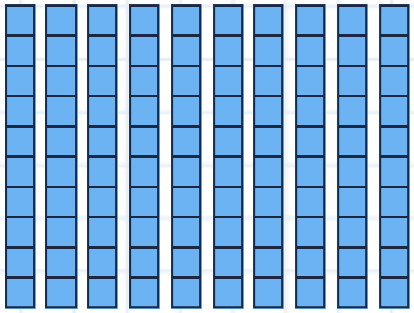
_____ tens _____ ones = _____



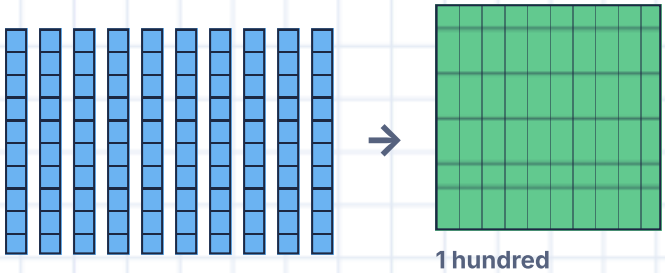
_____ tens _____ ones = _____

10 tens make 1 hundred

1 **Count by tens: 10, 20, 30, 40, 50, 60, 70, 80, 90, 100!**



2 **Stick 10 tens together into a big square. This is a HUNDRED.**

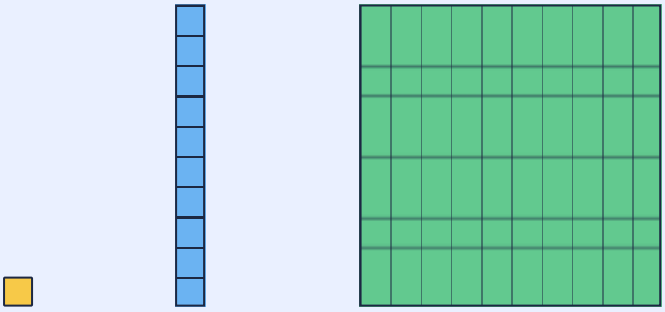


1 hundred

REMEMBER

10 tens = 1 hundred = 100 ones

ONE, TEN, HUNDRED



1 one = 1 | 1 ten = 10 | 1 hundred = 100

The **hundred** is the biggest. The **one** is the smallest.

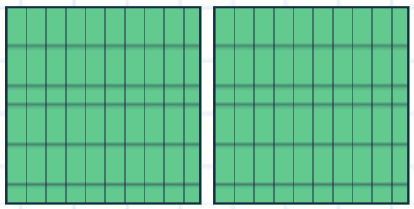
The number house

Every number lives in a house with rooms. Start from the **right**: **Ones**, then **Tens**, then **Hundreds**.

Hundreds	Tens	Ones

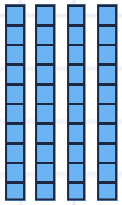
Let us build 243.

1 **2 hundreds go into the Hundreds room.**



Hundreds	Tens	Ones
2		

2 **4 tens go into the Tens room.**



Hundreds	Tens	Ones
2	4	

3 **3 ones go into the Ones room.**



Hundreds	Tens	Ones
2	4	3

Read it: **two hundred forty-three**.

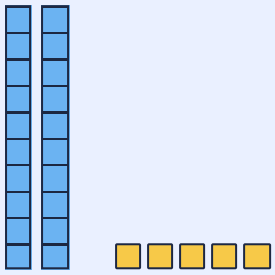
REMEMBER

Count the rooms from the RIGHT: Ones → Tens → Hundreds. The Ones room is always on the right.

The place matters!

25 and 52 have the same digits. Let us build both.

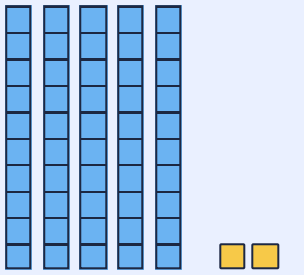
25 **2** **5**



2 tens 5 ones

The 2 is in the **tens** room. It means 2 tens = **20**.

52 **5** **2**



5 tens 2 ones

The 2 is in the **ones** room. It means 2 ones = **2**.

REMEMBER

Same digit 2, but a different value, because its **place** changed.

Place value = what a digit is worth in its place

YOUR TURN

In **36**, the 3 means _____ tens = _____. In **63**, the 3 means _____ ones = _____.

Face value

Face value is just the digit itself, the way it looks. Its own **face**!

THINK OF YOUR FACE

Sit in the front row: your face is the same. Sit in the back row: your face is **still the same**! A digit's face never changes either.

Number	Digit 7	Face value of 7
7	ones room	7
72	tens room	7
725	hundreds room	7

REMEMBER

Face value = the digit itself

It does not care about the place. It never changes.

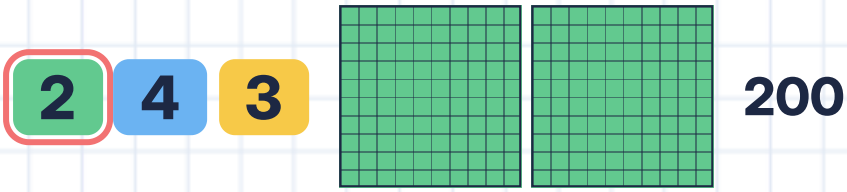
YOUR TURN

Face value of 4 in 648: _____ Face value of 9 in 197: _____ Face value of 1 in 1: _____

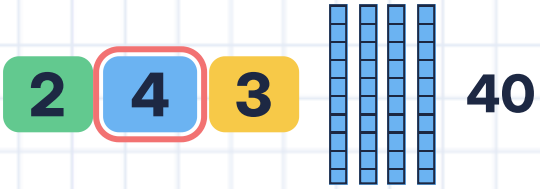
Place value

Place value tells how much a digit is worth in its room. Look at **243**.

1. **The 2 is in the hundreds room → 2 hundreds = 200.**



2. **The 4 is in the tens room → 4 tens = 40.**



3. **The 3 is in the ones room → 3 ones = 3.**



MAGIC TRICK

Write the digit, then put **one zero for every room after it**.

2 _ _ → 200 4 _ → 40 3 → 3

YOUR TURN

In **587**: place value of 5 = _____ · of 8 = _____ · of 7 = _____

Face value vs place value

Let us put them side by side for **365**.

Digit	Room	Face value	Place value
3	Hundreds	3	300
6	Tens	6	60
5	Ones	5	5

REMEMBER

Face value = the digit. **Place value** = the digit × its room.

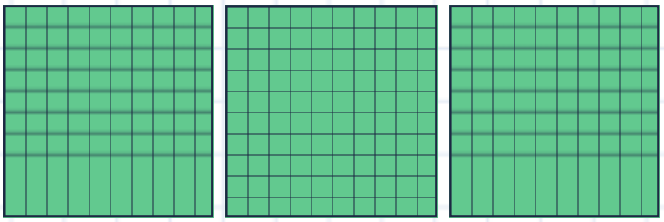
In the **ones room**, face value and place value are **always the same**.

YOUR TURN: FILL THE TABLE FOR 478

Digit	Room	Face value	Place value
4	_____	_____	_____
7	_____	_____	_____
8	_____	_____	_____

The Zero Hero

Look at **305**: 3 hundreds, **no tens**, 5 ones.



3 hundreds



5 ones

Hundreds	Tens	Ones
3	0	5

The **zero** sits in the tens room to keep it safe. It holds the place.

WHAT IF WE FORGET THE ZERO?

305 without the zero becomes **35**. That is a totally different number: thirty-five!

3 0 5 three hundred five ≠ **3 5** thirty-five

REMEMBER

Face value of 0 = **0**. Place value of 0 = **0**. Zero is always worth zero, in any room.

Zero holds the place.

YOUR TURN

1. 2 hundreds, 0 tens, 7 ones = _____ 2. 4 tens, 0 ones = _____

Expanded form

We can stretch a number out into its place values and add them. This is called **expanded form**.

1. **243 = 200 + 40 + 3**

   200 + 40 + 3

2. **586 = 500 + 80 + 6**

   500 + 80 + 6

3. **Backwards: 400 + 70 + 2 = 472**

   Put each digit in its room: 4 hundreds, 7 tens, 2 ones.

4. **Tricky: 600 + 9 = 609**

   No tens, so the Zero Hero sits in the tens room.

YOUR TURN

1. 352 = _____ + _____ + _____
2. 718 = _____ + _____ + _____
3. 300 + 20 + 4 = _____
4. 800 + 5 = _____

REMEMBER

Place value golden rules

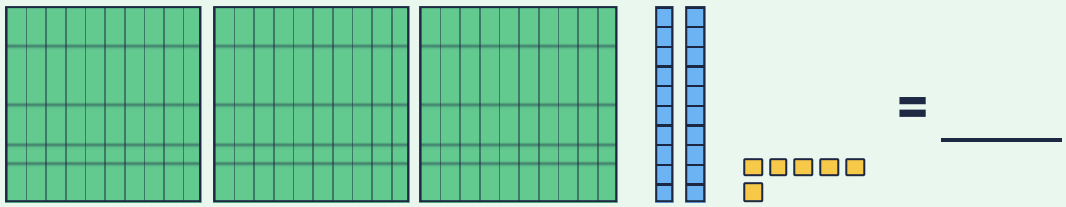
1. There are only 10 digits: 0 to 9.
2. 10 ones = 1 ten. 10 tens = 1 hundred.
3. Rooms from the right: Ones → Tens → Hundreds.
4. Face value = the digit itself. It never changes.
5. Place value depends on the room. 4 in the tens room = 40.
6. Zero holds the place. Its value is always 0.

QUICK CHECK

In 462...	Face value	Place value
4	4	400
6	6	60
2	2	2

Your turn!

1. WHAT NUMBER DO THE BLOCKS SHOW?



= _____

2. WRITE 476 IN THE NUMBER HOUSE

Hundreds	Tens	Ones

3. FACE VALUE AND PLACE VALUE

In **457**, the 5: face value _____, place value _____

In **832**, the 8: face value _____, place value _____

In **619**, the 9: face value _____, place value _____

In **508**, the 0: face value _____, place value _____

4. EXPANDED FORM

a) 274 = _____ + _____ + _____ b) 700 + 40 + 1 = _____ c) 900 + 3 = _____

5. CIRCLE THE RIGHT ANSWER

a) Which digit is in the tens place in 386? 3 / 8 / 6

b) Which number has 5 tens? 152 / 125 / 512

c) Which is bigger? 47 / 74

Answers

Step 1	10 digits · biggest 9, smallest 0 · 38 and 83
Step 2	4 ones · 9 ones
Step 3	3 tens 5 ones = 35 · 1 ten 8 ones = 18
Step 6	36: 3 tens = 30 · 63: 3 ones = 3
Step 7	4 · 9 · 1
Step 8	500 · 80 · 7
Step 9 (478)	4: hundreds, 4, 400 · 7: tens, 7, 70 · 8: ones, 8, 8
Step 10	207 · 40
Step 11	300 + 50 + 2 · 700 + 10 + 8 · 324 · 805
Practice 1	326
Practice 2	Hundreds 4 · Tens 7 · Ones 6
Practice 3	5 and 50 · 8 and 800 · 9 and 9 · 0 and 0
Practice 4	200 + 70 + 4 · 741 · 903
Practice 5	a) 8 · b) 152 · c) 74